

# TCP REDESIGN

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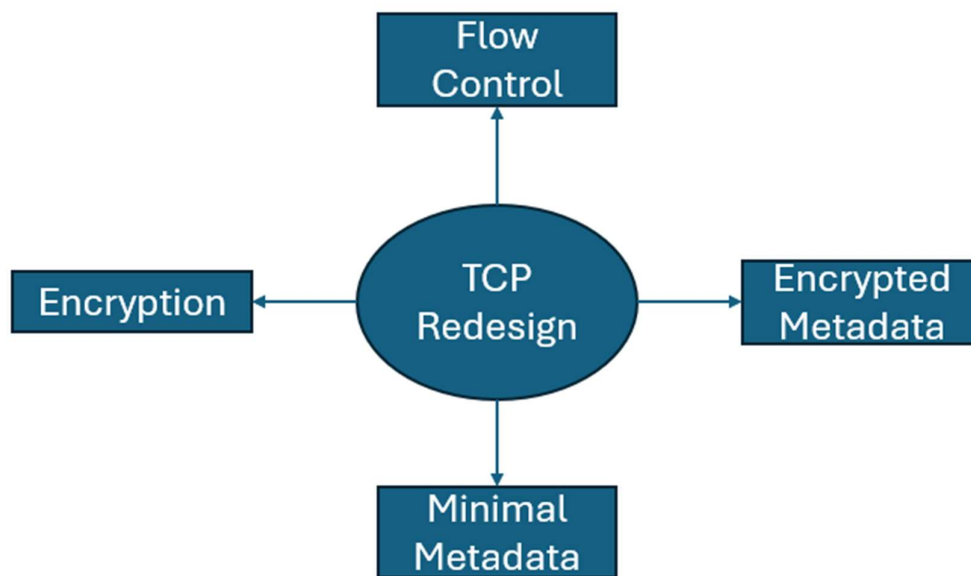
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## Introduction

This TCP redesign is designed to make TCP's foundation more secure and reliable compared to the current protocol. This redesign was created as TCP is one of the most fundamental network protocols, we currently have. This redesign was made to address the current problems in TCP specifically within reliability, security and privacy. This TCP redesign does not change how TCP works fundamentally using flags and other acknowledgment protocols. This TCP redesign instead will focus on features such as encryption, flow control, error-correction, limiting, and encrypting metadata.

This TCP redesign cannot promise perfect security, privacy and reliability as no network protocol can. However, this TCP redesign can promise to try providing a more secure and reliable protocol. This TCP redesign is targeted for all systems that currently use the TCP while the technical specifications are made for network engineers to use for the actual implementation of this protocol. Make sure to adapt the concepts or technical specifications based on resources and implementation.



The spider diagram above shows the 4 main concepts involved in this TCP redesign. For those, who may not know what flow control it is controlling the number of packets sent to destination at a time. This is important to not overflow the system and experience packet loss. You can find more information on the technical details of these concepts in the section below.

## **TCP Redesign Security and Privacy**

- This TCP redesign has 2 security and privacy features
- The first security/privacy feature is built-in encryption.
- TCP will use built-in encryption to secure the data within packets from interception or observation.
- This built-in encryption can use DH, PKI, PAKE that encrypts the data at the source and only decrypts it once the destination has been reached.
- DH can use ephemeral keys to prevent a single point of failure.
- This removes the need for HTTPS as all connections are encrypted meaning there is no need for certificates or TLS.
- This is beneficial for security as attackers cannot target CAs and exploit the differences between HTTPS and HTTP.
- The second security/privacy feature is metadata handling
- There are two options for handling metadata within TCP.
- The first option is encrypting metadata
- This is where metadata is wrapped within the encrypted payload to protect it from attackers.
- TCP currently has the metadata source port, destination port, sequence number, acknowledgement number, data offset, reserved, flags, window size, checksum, urgent pointer, options, padding
- Non-essential metadata such as data offset, checksum, urgent pointer, reserved bits, padding can be encrypted until it reaches the destination.
- Metadata such as source port, destination port, flags, sequence number, acknowledgement number, window size and TCP options must stay in the header.
- This is because they are needed for the middleman devices and basic routing capabilities.
- This is beneficial for privacy as an attacker cannot identify all the metadata within a packet that could be used to identify the data inside the packets.
- The second option is limiting metadata.
- Certain metadata can be limited such as TCP options with timestamps or what's needed based on real-world systems.
- This is important because TCP does not to collect metadata that isn't used for functionality.
- It is important to remember that usability must always be compatible with security otherwise security is pointless.
- You may adapt the security and privacy features in this section as needed to real-life applications.

## **TCP Flow Control and Error Correction Systems**

- TCP must be able to handle the flow of packets heading to a destination and handle any errors that may occur with the transmission.
- I will first begin by discussing the flow control system.
- This flow control system will be client-driven and proactive compared to TCP's current model of reactive and guesswork control.
- Once the connection is established, the destination/client sends out the packet rate that is useable for this.
- The client calculates this packet rate by measuring network bandwidth, current load (CPU usage, memory usage), latency, device capabilities, packet loss and application needs.
- TCP sends packets at the rate the client defined instead of guessing which ensures that the client isn't overloaded.
- If the client becomes overloaded, they can send a new packet rate to slow down the number of packets sent.
- This benefits performance as the client isn't overloaded and the packets are sent as per the needs of the system.
- I will now discuss the error correction system.
- This error-correction system will be proactive and reliable compared to TCP's current model which is reactive and relies on failure happening.
- Before transmission , TCP uses the checksums to check the integrity of data to ensure no data corruption has occurred.
- If corruption is detected, the data is repaired at the source either by recreating the packet or using error-correcting tools directly at the source.
- The data is then encrypted using the handshake protocol and sent.
- During transmission, data is encrypted meaning corruption is less likely as modern encryption algorithms have built-in integrity checks.
- Once the data has reached the destination, the client verifies the checksums to ensure that data isn't corrupted.
- If the checksum is valid, the data is accepted and the client sends a positive acknowledgement flag.
- If the checksum is invalid, the data is dropped and the client send a negative acknowledgement flag.
- TCP then resends the data to respond to the retransmission request, and the same checksum process is repeated.
- After multiple transmission failures, the connection will be dropped and then an automatic reconnection is attempted.
- This is to ensure that the issue wasn't a connectivity issue and not a direct packet issue.
- There will be a maximum of 2-3 reconnection attempts before the user is notified and the connection is terminated permanently.

## **TCP Redesign Trade-Offs**

- **This TCP redesign has trade-offs just like any other system I have created.**
- **There may also be further implementation challenges as each system uses TCP differently, but the core principles remain the same.**
- **I have identified 5 main trade-offs with this redesign.**
- **The first trade-off is performance.**
- **This trade-off exists because the encryption overhead and the extra checks can slow down overall performance within the TCP protocol.**
- **This means that systems use TCP may experience slower performance than the current TCP protocol.**
- **This trade-off can be managed by using UDP for high-speed applications, using fast encryption algorithms and accepting that performance won't be extremely fast.**
- **The second trade-off is complexity.**
- **This trade-off exists because the systems suggested and detailed in this TCP redesign can be difficult to implement.**
- **This means that network engineers may struggle to implement these systems efficiently.**
- **This trade-off can be managed by clear allocation of responsibility and duties based on skills, structured project management and testing of features.**
- **The third trade-off is time-consumption.**
- **This trade-off exists because this will likely take network engineers years to implement fully.**
- **This means that this will require constant and consistent effort for implementation and testing to ensure effectiveness.**
- **This trade-off can be managed by using project management, using existing algorithms and systems and rotating schedules to avoid burnouts.**
- **The fourth trade-off is compatibility.**
- **This trade-off exists because this redesign of the TCP protocol won't be compatible for every single system.**
- **This means that legacy systems will not have the more reliable and secure version of TCP.**
- **This trade-off can be managed by updating the TCP protocol gradually, testing features in simulations before real usage and avoiding the creation of a completely new TCP.**
- **The fifth and final trade-off is adoption.**
- **This trade-off exists because TCP is a fundamental protocol meaning that adopting a new version of it would take long.**
- **This trade-off can be managed by showing the effectiveness of the protocol and accepting that not everyone will adopt it immediately.**